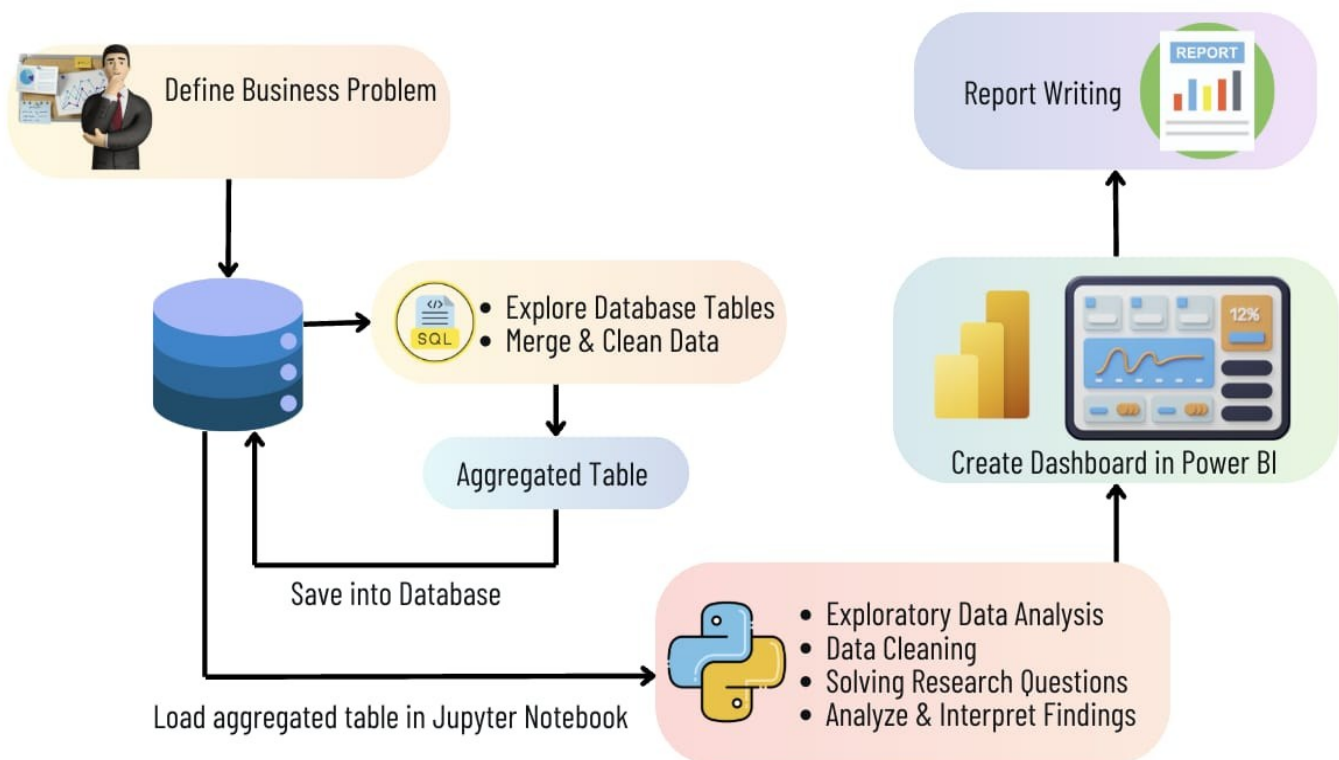


Vendor Performance Analysis Project



Business Problem

Efficient inventory and sales management are critical for optimizing profitability and meeting business objectives. Many companies struggle with inventory issues caused by poor vendor performance, which often results in **overstocking** or **understocking** of products.

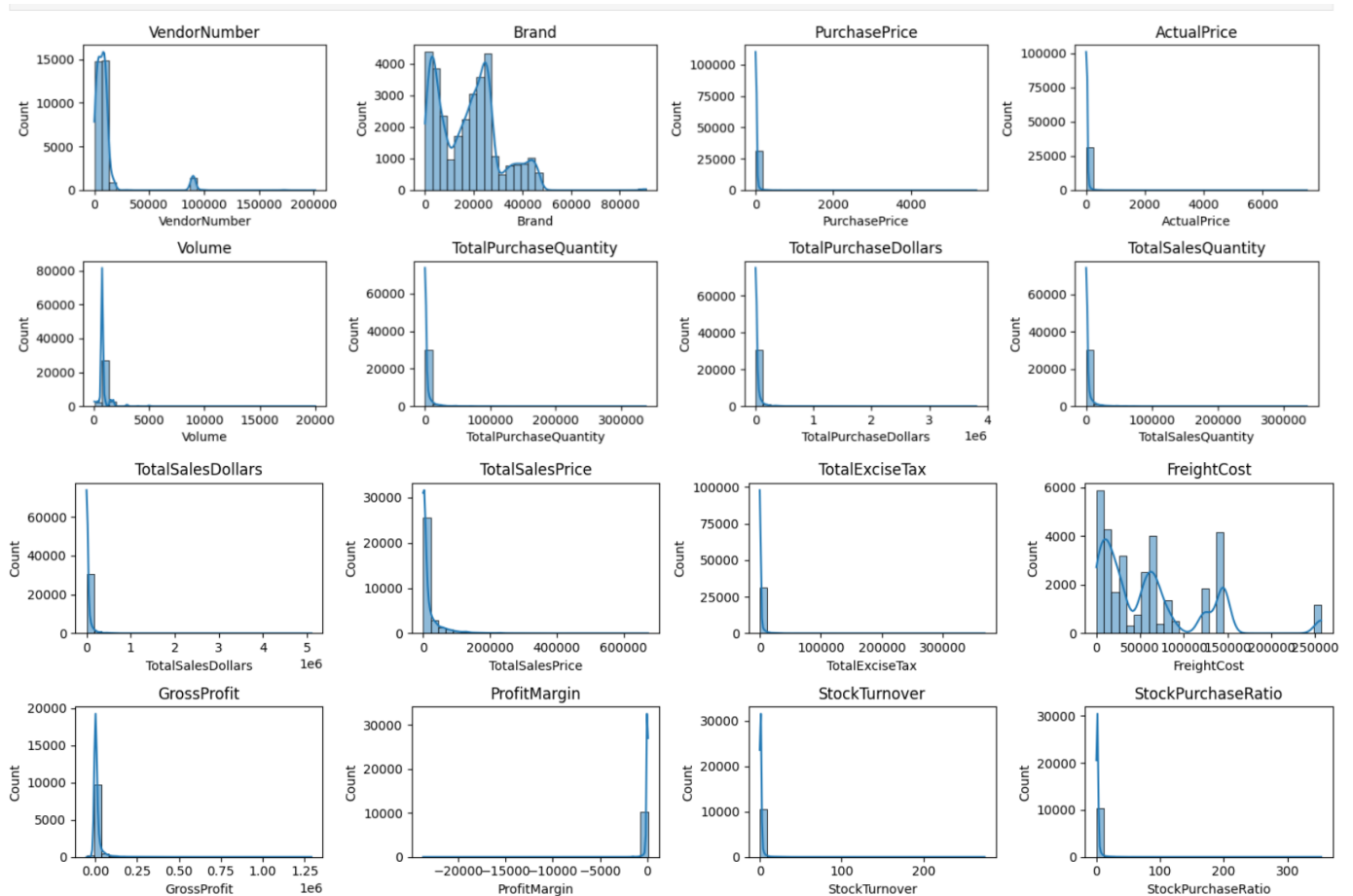
A deep understanding of vendor performance is essential because it enables better **promotional** and **pricing decisions**. Addressing these challenges involves:

- **Minimizing products** that contribute little to sales and gross profit.
- **Analyzing top vendors** to assess their profitability.
- **Identifying key vendors** that are vital to overall business health and strengthening relationships with them.
- **Developing a Vendor Performance Dashboard** that supports both long-term strategic decisions and short-term operational actions.

Exploratory Data Analysis insight

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	19.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-inf	NaN	-inf	13.324515	50.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02



Negative & Zero Values

- **Gross Profit:** The minimum value is $-52,002.78$, which suggests very high costs or heavy discounts. This likely comes from selling products at prices lower than their purchase costs.
- **Profit Margin:** The minimum is $-\infty$, meaning there are cases where revenue is zero or even less than total cost, leading to extreme negative margins.
- **Total Sales Quantity & Sales Dollars:** Some products show zero sales, indicating they were purchased but never sold. These are slow-moving or obsolete items, creating inventory inefficiencies.

Outliers Detected by High Standard Deviations

- **Purchase & Actual Prices:** Maximum values (5,681.81 and 7,499.99) are far higher than the mean (24.39 and 35.64). This points to premium product offerings compared to the average.
- **Freight Cost:** Extreme variation from **0.09 to 257,032.07** suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.
- **Stock Turnover:** Ranges from **0 to 274.5**, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than **1** indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

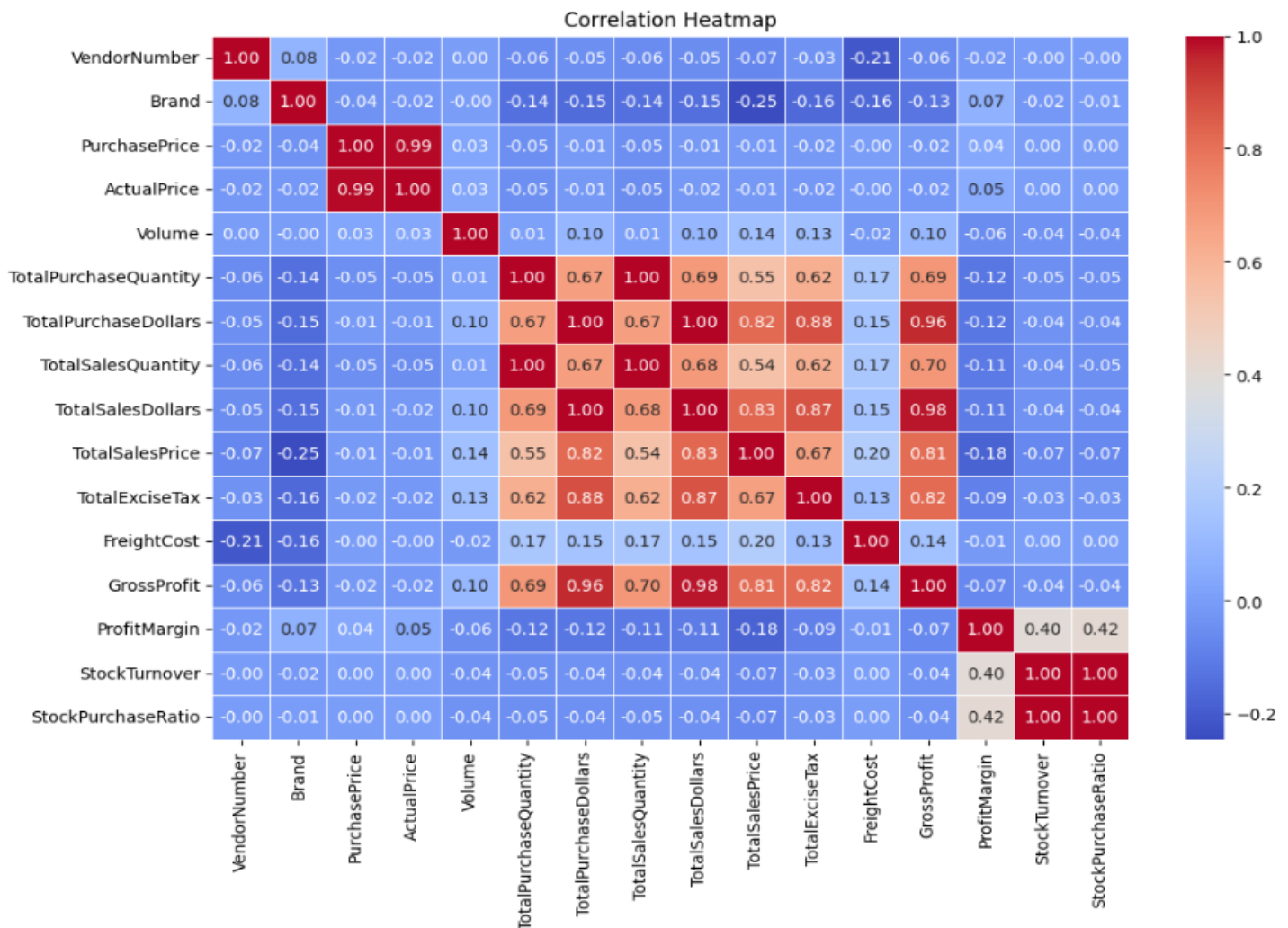
Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- **Gross Profit ≤ 0** (to exclude transactions leading to losses).
- **Profit Margin ≤ 0** (to ensure analysis focuses on profitable transactions).

- **Total Sales Quantity = 0** (to eliminate inventory that was never sold).

Correlation Analysis Insights



- **Purchase Price vs. Total Sales Dollars & Gross Profit:** Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.
- **Total Purchase Quantity vs. Total Sales Quantity:** Strong correlation (0.999), confirming efficient inventory turnover.
- **Profit Margin vs. Total Sales Price:** Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

- **Stock Turnover vs. Gross Profit & Profit Margin:** Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Research Question and Key finds

1.Brands for Promotional or Pricing Adjustments

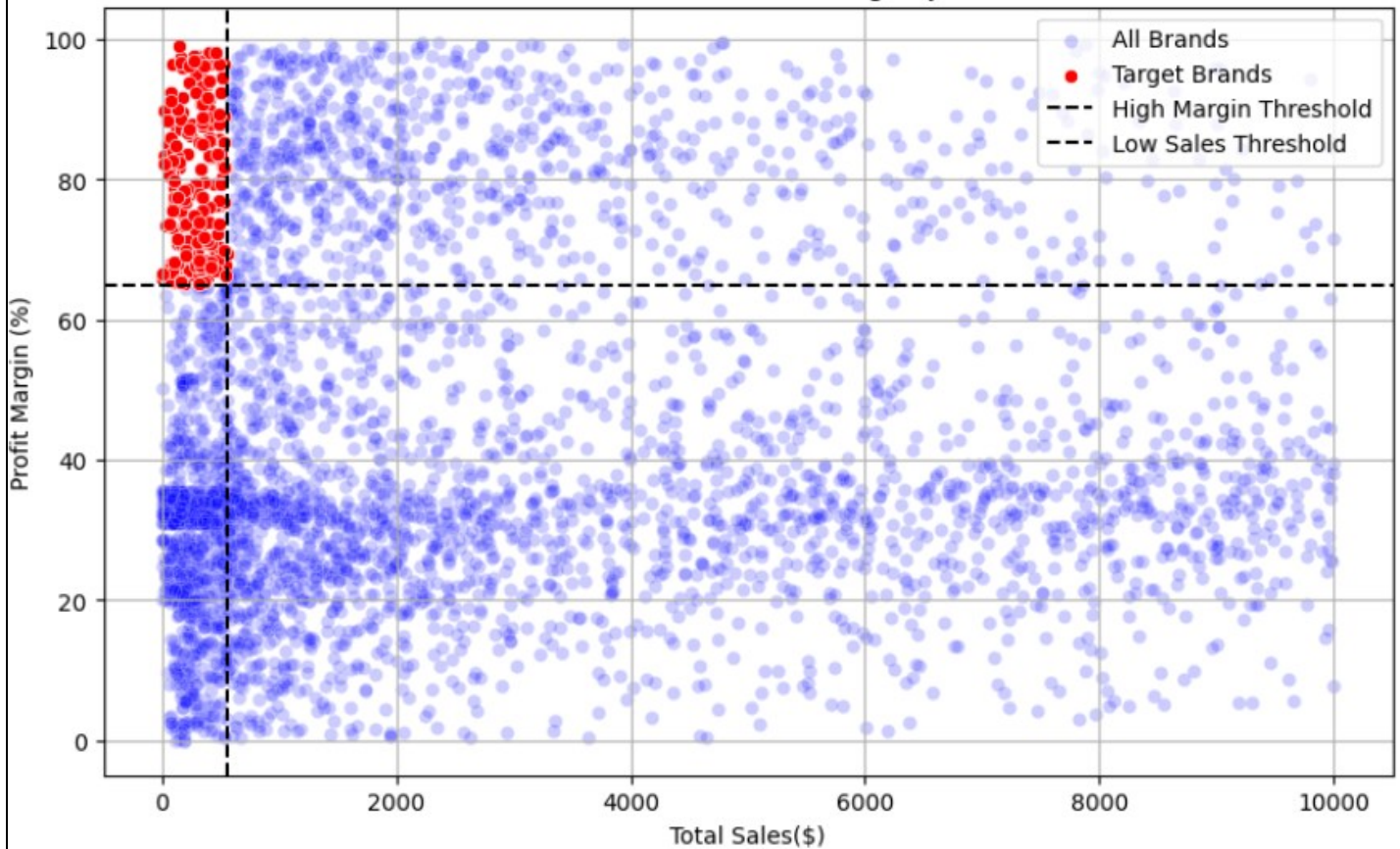
Brands with low Sales but High Profit Margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournal 10	558.87	69.434752

198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefits from targeted marketing , promotions , or price optimizations to increase volume without compromising profitability.

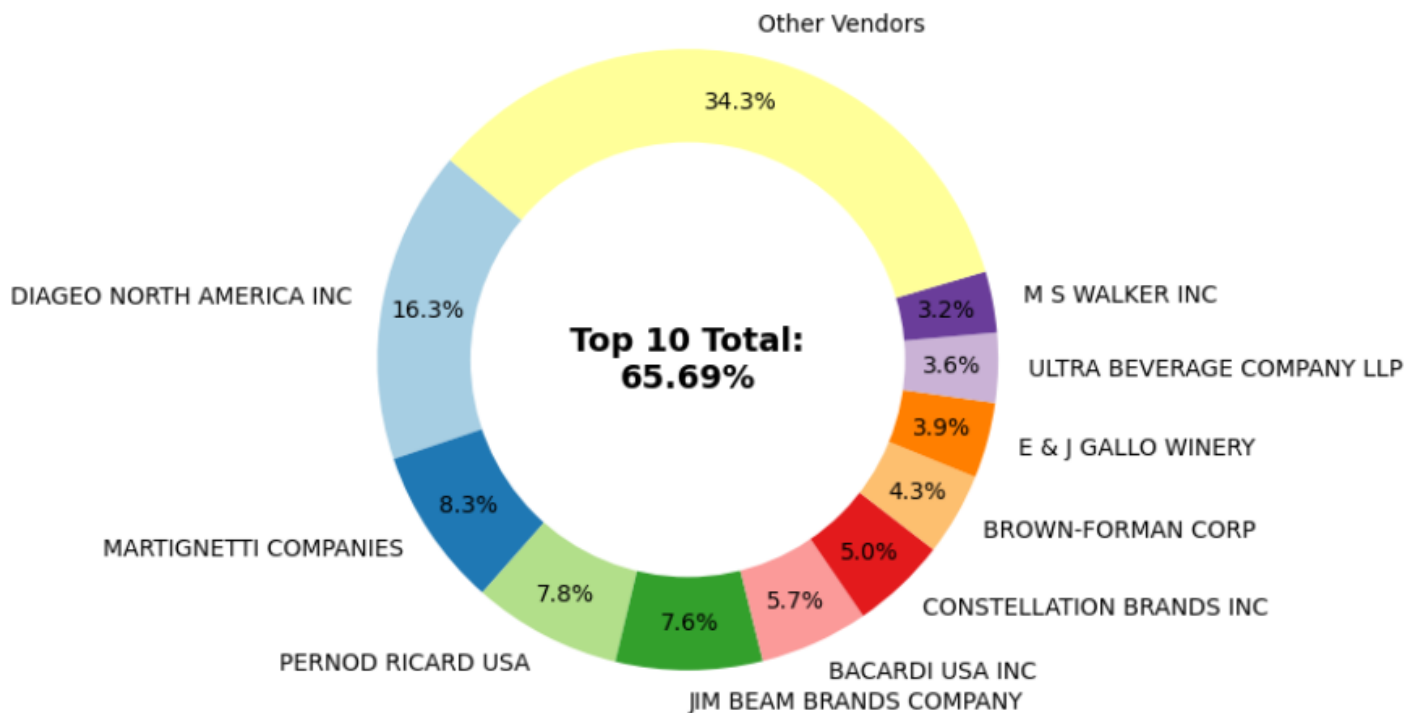
Brands for Promotional or Pricing Adjustments



2. Top vendors by sales and Purchase Contribution

The top 10 vendors contribute 65.69% of total purchase, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may indicate risks such as supply chain disruptions, indicating a need for diversification.

Top 10 Vendor's Purchase Contribution(%)



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost (\$10.78 per unit vs . higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintain profitability.

Order Size	Unit Purchase Price
Small	39.068186
Medium	15.486414
Large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital:\$2.71M

Slow-moving inventory increase storage costs, reduces cash flow efficiency , and affects overall profitability .

Identifying vendors with low inventory turnover enables better stock management , minimizing financial strain .

Low Inventory Turnover Vendors

Vendor Name	Stock Turnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WAI POI F MTN VIEW WINFRY	0.820548

Top Vendors by Unsold Inventory Value

Vendor Name	Unsold Inventory Value
DIAGEO NORTH AMERICA INC	\$722.21K
JIM BEAM BRANDS COMPANY	\$554.67K
PERNOD RICARD USA	\$470.63K
WILLIAM GRANT & SONS INC	\$401.96K
E & J GALLO WINERY	\$228.28K
SAZERAC CO INC	\$198.44K
BROWN-FORMAN CORP	\$177.73K
CONSTELLATION BRANDS INC	\$133.62K
MOET HENNESSY USA INC	\$126.48K
REMY COINTREAU USA INC	\$118.60K

5. Profit Margin Comparison: High vs. Low-Performing

Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), **Mean:** 31.17%

Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), **Mean:** 41.55%

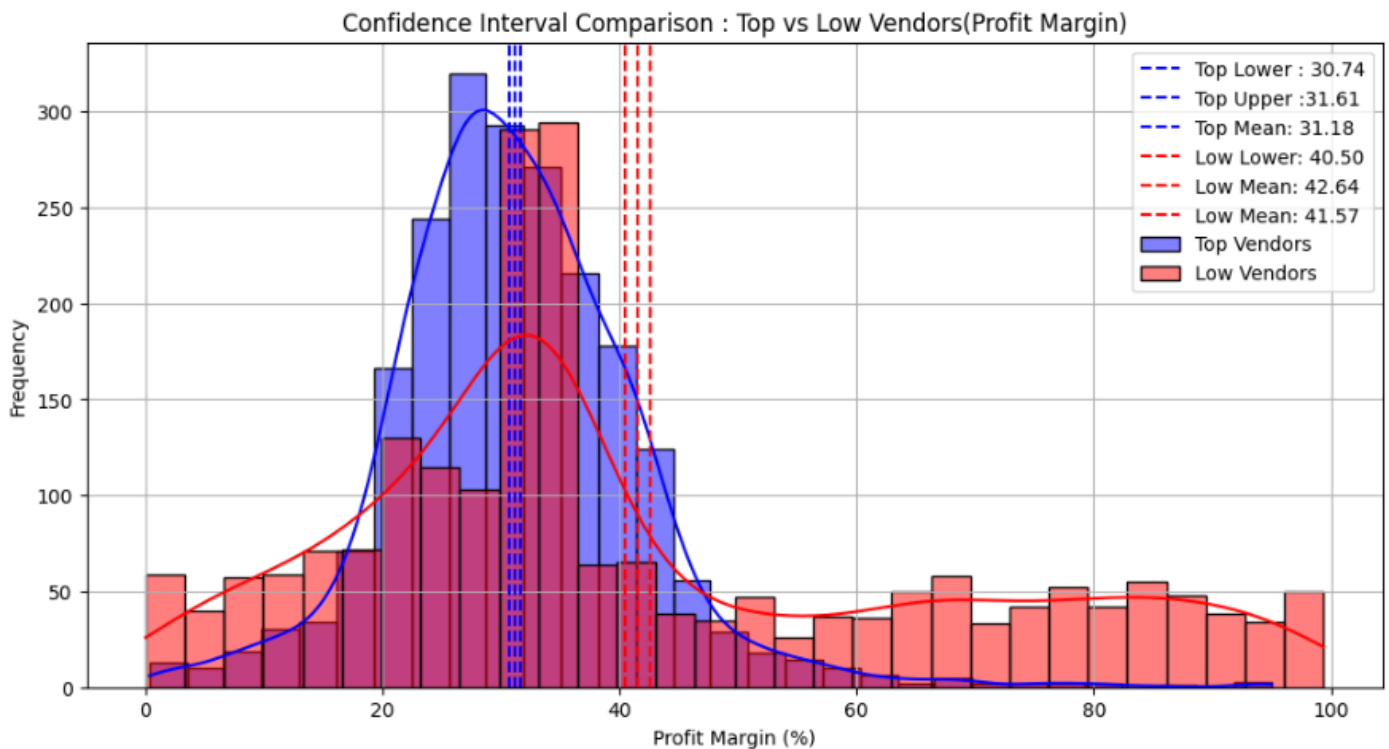
Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- **Top-performing vendors:** Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- **Low – performing vendors:** Improve marketing efforts, optimize pricing strategies , and enhance distribution networks.

Top Vendors 95% CI : (30.74, 31.61), mean: 31.18

Low Vendors 95% CI : (40.50, 42.64), mean: 41.57



6. Statstical Validation of Profit Margin Differences

Hypothesis Testing:

H₀ (Null Hypothesis): No significant difference in profit margins between top and low-performing vendors.

H₁ (Alternative Hypothesis): A significant difference exists in profit margins between the two vendor groups.

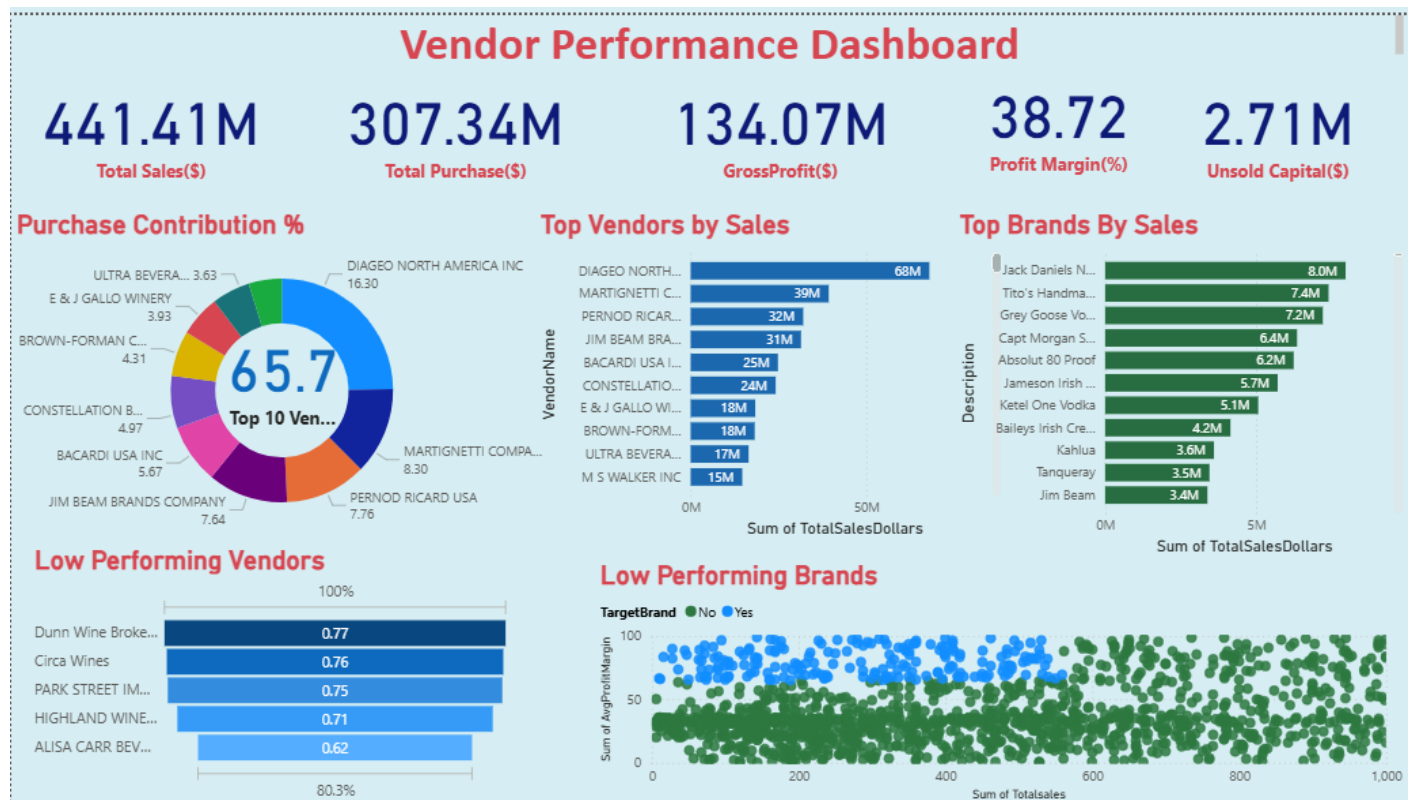
Result: The null hypothesis is rejected, confirming that the two groups operate under distinctly different profitability models.

Implication: High-margin vendors may benefit from better pricing strategies, while the top-selling vendors could focus on cost efficiency.

Final Recommendations

- Re-evaluate pricing for low – sales, high-margin brands to boost sales volume without sacrificing profitability
- Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.
- Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.
- Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.
- Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.
- By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency.

Vendor Performance Dashboard Insights



1. Overall Business Performance

- Total Sales: \$441.41M
- Total Purchases: \$307.34M
- Gross Profit: \$134.07M
- Profit Margin: 38.72%
- Unsold Capital: \$2.71M

Insight:

- The company generated **\$441.41M** in sales with a healthy **38.72% profit margin**, resulting in **\$134.07M gross profit**.
- **Unsold inventory worth \$2.71M** indicates capital tied up in slow-moving stock and presents an opportunity for inventory optimization.

2.Purchase Contribution (%)

Business Recommendation

- Diversify the supplier base to reduce procurement risk.
- Strengthen relationships with high-performing vendors while evaluating underperforming suppliers.

Insight:

- The **Top 10 vendors contribute approximately 65.7%** of total purchases.
- **Diageo North America Inc.** is the largest supplier, contributing **16.3%** of purchases.
- The remaining vendors contribute significantly less, indicating moderate dependence on a few major suppliers.

3.Top Vendors by Sales

Highest Performing Vendors

Vendor	Sales
DIAGEO NORTH...	68M
MARTIGNETTI...	39M
PERNOD RICARD	32M
JIM BEAM	31M
BACARDI USA	25M

Insight:

- **Diageo North America** generates nearly **twice the sales** of the second-highest vendor.
- Sales are concentrated among a few key vendors, making them strategically important.

4. Top Brands by Sales

Leading Brands

- Jack Daniel's – **8.0M**
- Tito's Handmade – **7.4M**
- Grey Goose Vodka – **7.2M**
- Captain Morgan – **6.4M**
- Absolut 80 Proof – **6.2M**

Insight:

- Premium spirit brands dominate sales.
- These brands should receive continued marketing and inventory support to maximize revenue.

5. Low Performing Vendors

Lowest performing vendors include

- Dunn Wine Brokers
- Circa Wines
- Park Street Imports
- Highland Wine
- Alisa Carr Beverages

Insight:

- These vendors have performance scores between **0.62 and 0.77**, indicating lower sales contributions.

- Their product mix, pricing, or demand should be reviewed to improve performance.

6. Low Performing Brands

Business Recommendation

- Reduce purchase quantities for consistently slow-moving brands.
- Introduce promotional campaigns or discounts to clear excess inventory.
- Focus procurement on brands with higher turnover.

Insight

- The scatter plot shows a large number of brands with **low sales despite adequate purchase volumes**.
- Several brands appear overstocked, suggesting inefficient inventory

7. Overall Business Insights

- Generated **\$441.41M** in revenue with a strong **38.72% profit margin**.
- **Top 10 vendors account for 65.7%** of purchases, indicating supplier concentration.
- **Diageo North America** is the leading vendor with **\$68M** in sales.
- **Jack Daniel's** is the highest-selling brand, generating **\$8M** in sales.
- **\$2.71M** in unsold inventory highlights an opportunity to optimize stock levels.
- Several vendors and brands consistently underperform, suggesting the need for better supplier evaluation and inventory optimization strategies.
- Sales are concentrated among a few high-performing vendors and brands, while many products exhibit low turnover, emphasizing the importance of demand-driven purchasing.

These insights are suitable for inclusion in a resume, portfolio, or project presentation because they focus on business impact rather than merely describing the charts.